A1

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5117
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage: **40%**

QUESTIONS: **Total Marks: 50**

Annexure A

Analytical Problem Solving

Code of Conduct:

I K·S·Dinesh(192324185) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Dinesh

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis

Given p=11, q=13

a. Compute n $\phi(n)$\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation

Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study

Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i.Calculate total encryption time for each algorithm
ii.Analyze trade-offs between security and performance
iii.Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis

Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison
Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganize d steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1) Block Cipher Mode Evaluation:

Plaintext Blocks $P_1 = 1010$

$P_2 = 1010$

$P_3 = 1111$

$P_4 = 1010$

Key $K = 1100$

initialization vector (IV) : 0000

a) Encryption using ECB mode:

Formula:

$$C_i = E_K(P_i) = P_i \oplus K$$

Block	Plaintext	Key	Ciphertext
P_1	1010	1100	0110
P_2	1010	1100	0110
P_3	1111	1100	0011
P_4	1010	1100	0110

ECB Ciphertext:

0110, 0110, 0011, 0110

b) Encryption using CBC mode:

$$C_1 = E_K(P_1 \oplus IV)$$

$$C_i = E_K(P_i \oplus C_{i-1})$$

Block 1:

$C_1 = (1010 \oplus 0000) \oplus 1100 = 0110$

Block 2:

$C_2 = (1010 \oplus 0110) \oplus 1100 = 0000$

Block 3:

$C_3 = (1111 \oplus 0000) \oplus 1100 = 0011$

Block 4:

$C_4 = (1010 \oplus 0011) \oplus 1100 = 0101$

c) Compare and analyze:

Feature	ECB Mode	CBC Mode
• Ciphertext	0110, 0110, 0011, 0110	0110, 0000, 0011, 0101
• identical Plaintext blocks	Produce identical ciphertext	usually Produce different ciphertext.
• Pattern hiding	Poor	Better
• IV required	No	yes
• security	Less secure for Patterned data	Better Pattern Protection.

→ CBC mode hides repeated Plaintext Patterns.

Diffie - Hellman Key Exchange Analysis:

Given:

Prime number, $P = 23$

Generator, $g = 5$

user A's private key, $a = 6$

user B's Private key, $b = 15$

a) Compute the public key of user A:

$$A = g^a \bmod P$$

Substitute the values:

$$A = 5^6 \bmod 23$$

$$5^6 = 15625 \bmod 23 = 8$$

b) Compute the Public Key of user B:

$$B = g^b \bmod P$$

$$B = 5^{15} \bmod 23$$

using modular exponentiation:

$$5^{15} \bmod 23 = 19$$

Therefore, user B's Public Key = 19

c) Compute the Shared secret key:

$$K_A = B^a \bmod P$$

$$K_A = 19^6 \bmod 23 = 2$$

$K_B = A^b \bmod P$

$K_B = 8^{15} \bmod 23 = 2$

$K = 2$

d) Demostrate a man-in-the middle:

user A

↓↑

AHacker Eve

↓↑

user B

$E = 5^3 \bmod 23 = 10$

$K_{AE} = 10^6 \bmod 23 = 6$

$K_{BE} = 10^{15} \bmod 23 = 19$

e) Suggest a secure method to prevent the attack:

1. Digital Signatures: Sign the exchanged Public keys to verify the authenticity.

2. Digital Certificates (PKI): use Certificates issued by a trusted Certificate Authority to authenticate users.

3. Authenticated Diffie-Hellman: Combine key exchange with authentication as TLS.

4. Verify Public Keys: Confirm that the received public